Contest Problem Set 12129

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Number Sense
12129

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Name _____

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1. $21 + 221 - 12 =$ _____.
2. The thousands digit of 942478 is _____.
3. $1021 + 281 =$ _____.
4. The product of 19 and 9 is _____.
5. $43 + 41 + 77 + 8 =$ _____.
6. $754 - 385 =$ _____.
7. $12 + 12 \times 12 =$ _____.
8. The remainder of $321 \div 4$ is _____.
9. $84 \div 3 =$ _____.
10. (estimate) $712 + 1068 + 2136 =$ _____.
11. $86 \times 11 =$ _____.
12. $818 - 409 - 112 =$ _____.
13. $14^2 =$ _____.
14. The remainder of $874 \div 9$ is _____.
15. $72 \times 55 =$ _____.
16. $252 + 315 + 693 =$ _____.
17. *CDXCVI* in Arabic numerals is _____.
18. $21 \times 39 =$ _____.
19. $18 \times 13 \times 5 =$ _____.
20. (estimate) $321 \times 399 =$ _____.
21. $\frac{42}{154}$ in lowest terms is _____ (fraction).
22. $107 \times 25 =$ _____.
23. $36 + 31 + 26 + 21 + 16 + 11 =$ _____.
24. The *GCD* of 75 and 165 is _____.
25. $2021 \div 43 =$ _____.
26. The greater of $\frac{6}{17}$ and $\frac{5}{14}$ is _____ (fraction).
27. $21^2 =$ _____.
28. If 1 ounce is equal to $\frac{1}{16}$ pounds, then 88 ounces is _____ pounds (decimal).
29. $1 + \frac{1}{12} - \frac{1}{3} =$ _____ (fraction).
30. (estimate) $36 \times 34 \times 11 =$ _____.
31. $99 \times 81 =$ _____.
32. The perimeter of a rectangle with width 8 and area 72 is _____.
33. 75% of 76 is _____.
34. 20 minutes and 21 seconds is _____ seconds.
35. $79 \times 71 =$ _____.
36. The number of multiples of 6 between 19 and 145 is _____.
37. The LCM of 75 and 165 is _____.
38. $155^2 =$ _____.
39. The greatest prime divisor of 2021 is _____.
40. (estimate) $733 \times 135 =$ _____.

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41. The remainder of $2021 \div 11$ is _____.

42. $88 \times 28 =$ _____.

43. $\frac{31}{37} \times 31 =$ _____ (mixed number).

44. The number of diagonals in a regular pentagon is _____.

45. $71^2 - 54^2 =$ _____.

46. $28 \times 11 + 16 \times 7 =$ _____.

47. The number of terms in the arithmetic sequence $7, 16, 25, \ldots, 277$ is _____.

48. $391 \times 101 =$ _____.

49. $5^6 \times 2^3 =$ _____.

50. (estimate) $\sqrt{2345678} =$ _____.

51. $\sqrt{23104} =$ _____.

52. 21_{16} in base 2 is _____ $_2$.

53. The area of an isosceles right triangle with hypotenuse of length 6 is _____.

54. $126 \times 103 =$ _____.

55. The median of the list $64, 2, 16, 8, 32, 4$ is _____.

56. $625 \times 13 =$ _____.

57. $31\frac{1}{4}\%$ of 96 is _____.

58. $96 \times 78 =$ _____.

59. The perimeter of a rectangle with a diagonal of length 17 and a width of 8 is _____.

60. (estimate) $857142 \times 9 =$ _____.

61. Four fair coins are flipped. The probability at least two show heads is _____ (fraction).

62. $27 \times 444 =$ _____.

63. The number 261 written in base 5 is _____ $_5$.

64. $21^3 =$ _____.

65. The measure of an interior angle of a regular dodecagon is _____ $^\circ$.

66. $7\frac{2}{3} \times 5\frac{2}{3} =$ _____ (mixed number).

67. If $v = 47$, then $v^2 - 4v =$ _____.

68. $5 \times 7 \times 14 \times 19 =$ _____.

69. $0.1\overline{36} =$ _____ (fraction).

70. (estimate) $77^4 \div 10^4 =$ _____.

71. The diameter of a sphere with surface area 225π is _____.

72. $1029 \times 1031 =$ _____.

73. The sum of the positive whole number divisors of 2021 is _____.

74. $925^2 =$ _____.

75. If $9^{x-1} = 16$, then $3^{x+1} =$ _____.

76. $\frac{4}{5}$ of 22 is $\frac{11}{15}$ of _____.

77. $\sqrt{147} \times \sqrt{27} =$ _____.

78. $26 \times 75 \times 62 =$ _____.

79. The number of proper subsets of the set $\{H, O, M, E, R, U, N\}$ is _____.

80. (estimate) $\sqrt{2021 \times 20 \times 21} =$ _____.

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Sprint Round 12129

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Select E only if you cannot determine a uniquely correct answer between A, B, C, and D.

1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. Emily multiplied 99×123. What is the sum of the digits of Emily's result?

(A) 12 (B) 16 (C) 24 (D) 18 (E) Other

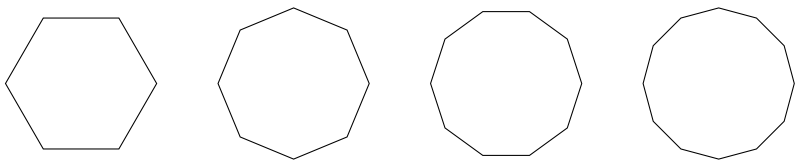
2. On a mathleague.org Sprint Round, a correct answer is worth 4 points, and an incorrect answer is worth -1 point. At the state competition, Andrew answered 16 questions correctly on the Sprint Round, and he answered 3 questions incorrectly. What was Andrew's score on the Sprint Round at the state competition?

(A) 61 (B) 64 (C) 65 (D) 67 (E) Other

3. How many even positive whole numbers less than 50 are also multiples of 3?

(A) 8 (B) 12 (C) 6 (D) 10 (E) Other

4. A hexagon, an octagon, a decagon, and a dodecagon are shown below. How many total sides are shown?



(A) 42 (B) 36 (C) 38 (D) 40 (E) Other

5. Alan, Beth, and Candace are walking into a haunted house at the carnival. Alan is scared, so he refuses to be the first person among the three friends to walk into the haunted house. In how many different orders can the three friends walk into the haunted house?

(A) 4 (B) 2 (C) 6 (D) 5 (E) Other

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6. A three-digit positive whole number is divisible by 17. The hundreds digit of the number is a 5 and the units digit of the number is an 8. What is the tens digit of the number?

(A) 5 (B) 7 (C) 8 (D) 9 (E) Other

7. Wesley has sixteen quarters and eleven dimes. Victoria has three more quarters than Wesley and five fewer dimes than Wesley. Quarters are coins worth 25 cents and dimes are coins worth 10 cents. What is the total value of Victoria's quarters and dimes?

(A) $5.35 (B) $5.25 (C) $5.55 (D) $5.45 (E) Other

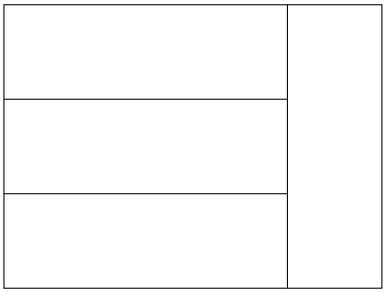
8. Aditya rode his bike to his friend's house that was four miles away. If Aditya rode at an average speed of twelve miles per hour, then how many minutes did it take Aditya to ride his bike to his friend's house?

(A) 15 (B) 30 (C) 24 (D) 20 (E) Other

9. How many positive whole numbers evenly divide both 60 and 90?

(A) 7 (B) 10 (C) 6 (D) 8 (E) Other

10. The figure below is comprised of four identical rectangles and has a total area of 300. What is the perimeter of the figure?



(A) 80 (B) 60 (C) 90 (D) 70 (E) Other

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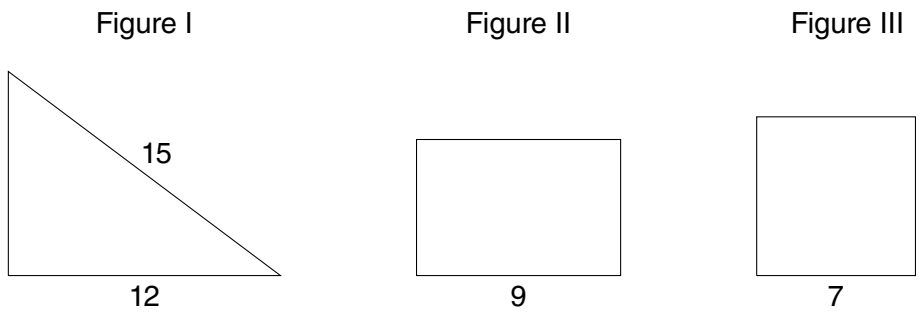
11. In the year 2021, January 1st was a Friday. What day of the week is December 1st, 2021?

(A) Tuesday (B) Friday (C) Thursday (D) Saturday (E) Other

12. At Crystal Lake Elementary School, the ratio of fourth graders to fifth graders is 5 : 7, and the ratio of third graders to fourth graders is 4 : 3. If there are 60 more fifth graders than fourth graders, then how many total students are in the third, fourth, and fifth grades at Crystal Lake Elementary School?

(A) 560 (B) 510 (C) 540 (D) 480 (E) Other

13. Shown below are three figures. Figure *I* is right triangle with a hypotenuse of length 15 and a leg of length 12. Figure *II* is a rectangle with perimeter 30 and width 9. Figure *III* is a square of side length 7. Which of the figures have area greater than or equal to the area of all the other figures?



(A) *II* & *III* (B) *I* & *II* (C) *II* only (D) *I* & *III* (E) Other

14. In a box of 120 holiday ornaments, 75% are green and the rest are red. Peter removes green ornaments from the box until only 60% of the ornaments in the box are green. How many green ornaments did Peter remove from the box?

(A) 40 (B) 30 (C) 55 (D) 45 (E) Other

15. A number is twice as far from 17 as it is from 23 on the number line. What is the sum of all possible values of the number?

(A) 29 (B) 32 (C) 21 (D) 50 (E) Other

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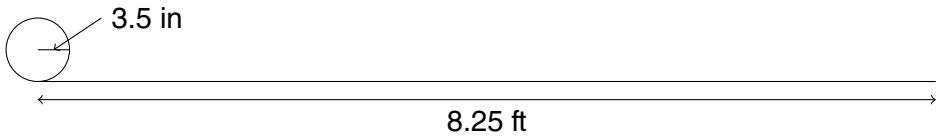
16. For lunch in the school cafeteria, students may select a main course of a hamburger, a vegetarian burger, or pizza. Additionally, students may select vegetables from among corn, green beans, carrots, or broccoli. Finally, students may select a dessert of either ice cream or cake. If a lunch consists of one main course, two different vegetables, and one desert, then how many different lunches are possible?

(A) 36 (B) 48 (C) 24 (D) 30 (E) Other

17. The median and mean of five positive whole numbers are equal. Four of the numbers are 28, 20, 4, and 12. What is the sum of all possible values for the fifth number?

(A) 48 (B) 36 (C) 56 (D) 52 (E) Other

18. A wheel of radius 3.5 inches rolls, without slipping, along a path that has a length of 8.25 feet. There are 12 inches in 1 foot. To the nearest tenth of a revolution, how many revolutions does the wheel make when rolling along the entire length of the path?



(A) 4.5 (B) 5.0 (C) 4.3 (D) 4.2 (E) Other

19. Alicia chose a positive whole number. When Alicia wrote the value of the whole number in base four, she used four digits. When Alicia wrote the value of the whole number is base six, she used three digits. How many different positive whole numbers could Alicia have chosen?

(A) 154 (B) 153 (C) 151 (D) 152 (E) Other

20. If $x \diamondsuit y = xy + x + y + 1$ and $x \heartsuit y = xy - x - y + 1$, then what is the value of $(38 \heartsuit 28) \diamondsuit (22 \heartsuit 20)$?

(A) 400000 (B) 399999 (C) 400001 (D) 400400 (E) Other

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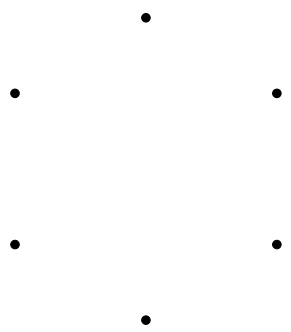
21. Andre wrote down a three-digit number with three different digits, none of which were zero. He then wrote down all other three-digit numbers that had the same three digits in a different order. Andre then summed all of the numbers he wrote down. Which of the following is a possible value for Andre's sum?

(A) 3107 (B) 3110 (C) 3108 (D) 3111 (E) Other

22. The number one trillion is a 1 followed by 12 zeroes. What is the smallest whole number power of 8 that is greater than one trillion?

(A) 8^{13} (B) 8^{15} (C) 8^{14} (D) 8^{12} (E) Other

23. The dots below are vertices of a regular hexagon. How many non-congruent convex polygons can be formed by using some or all of these dots as vertices of the polygon?



(A) 7 (B) 6 (C) 5 (D) 9 (E) Other

24. A soccer league has six teams. During the season, every team in the league plays every other team in the league twice. The standings of teams in the league are calculated using a point system where a win is worth 2 points, a tie is worth 1 point, and a loss is worth 0 points. At the end of the season, the bottom five teams in the league standings had earned 13, 12, 9, 7, and 2 points. How many points did the first place team in the league standings earn?

(A) 15 (B) 16 (C) 17 (D) 14 (E) Other

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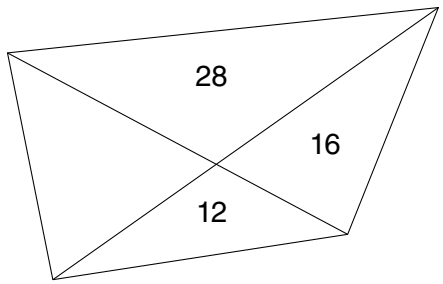
25. In the figure below, diagonals partition a quadrilateral into four triangles. Three of the triangles have areas of 12, 16, and 28, as labeled. What is the total area of the quadrilateral?



(A) 80 (B) 72 (C) 75 (D) 76 (E) Other

26. A number is the product of all positive whole number divisors of 18^5. What is the greatest power of 3 that evenly divides that number?

(A) 3^{330} (B) 3^{300} (C) 3^{280} (D) 3^{275} (E) Other

27. Michelle and Nala each have a bag containing three red and three blue marbles. Michelle draws two marbles from Nala's bag and then adds them to her bag. Nala then draws two marbles from Michelle's bag and adds them to her bag. After this occurs, what is the probability that Nala's bag contains three red marbles and three blue marbles?

(A) $\frac{15}{28}$ (B) $\frac{17}{35}$ (C) $\frac{19}{35}$ (D) $\frac{13}{28}$ (E) Other

28. Beginning at the units digit and moving leftward, the product 99! has a number of digits that are zero. What is the value of the first digit that is not zero?

(A) 2 (B) 8 (C) 4 (D) 1 (E) Other

29. Values x, y, and z, and a, b, and c exist such that $ax + by + cz = 1$, $a + bx + cy = z$, $b + cx + az = y$, $c + ay + bz = x$, and $a + b + c = 2021$. What is the sum of x, y, and z?

(A) 2021 (B) 2020 (C) 0 (D) -1 (E) Other

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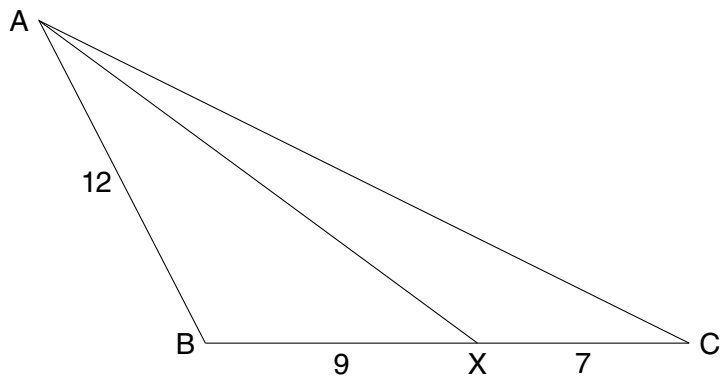
30. In triangle *ABC*, shown below, side *AB* has length 12, and point *X* is located on side *BC* so that the length of *BX* is 9 and the length of *XC* is 7. The perimeter of triangle *AXC* is 49. What is the square of the area of triangle *ABC*?



(A) 7160 (B) 7280 (C) 7360 (D) 7450 (E) Other

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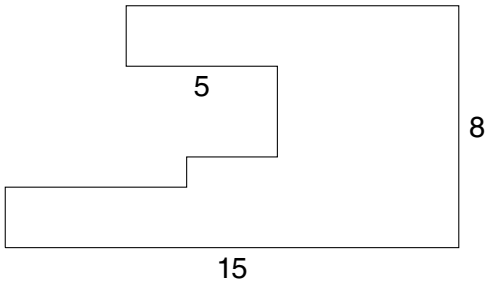
Problems 1 & 2

1. Jay has a part-time job at the local market to earn extra money. Jay works two hours after school every Tuesday and two hours after school every Thursday. He also works five hours every Saturday. If Jay is paid $9 per hour, then how many dollars does Jay earn in six weeks?

1.

2. In the figure below, all sides meet at right angles, and three side lengths measure 15, 8, and 5, as shown. What is the perimeter of the figure?

2.



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Problems 3 & 4

3. A prime number is a positive whole number that is only evenly divisible by itself and 1. For example, since 31 is only evenly divisible by 31 and 1, the whole number 31 is prime. How many whole numbers greater than 100 and less than 120 are prime?

3.

4. Julia has a number of beads and a number of jars. When Julia places five beads in each jar, she has five beads left over. When Julia leaves one jar empty and places six beads in each of the other jars, she has no beads left over. How many beads does Julia have?

4.

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Problems 5 & 6

5. Using each of the digits 9, 8, 6, and 0 exactly once, Collin created a four-digit positive whole number that is divisible by 11. How many different numbers could Collin have created?

5.

6. If the order of the numbers chosen does not matter, then how many different sets of three whole numbers greater than or equal to 1 and less than or equal to 15 have no consecutive whole numbers? For example, one such set is 5, 2, and 13, since 2 and 5 are not consecutive, 2 and 13 are not consecutive, and 5 and 13 are not consecutive.

6.

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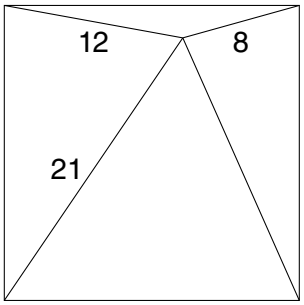
Problems 7 & 8

7. Kevin, Lara, Maria, and Nicole each chose a positive whole number. Kevin's number was less than Lara's number, Lara's number was less than Maria's number, and Maria's number was less than Nicole's number. The product of all four numbers was 72. What is the sum of all possible values of Nicole's number?

7.

8. A point is located in the interior of a square. As shown below, three of the distances from the point to the vertices of the square are 12, 8, and 21. What is the square of the distance from the point to the fourth vertex of the square?

8.



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Team Round
12129

School or Team

Name _____

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Name _____

Name _____

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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1. How many positive whole numbers less than 101 are multiples of 4 but are not multiples of 6?

2. Tickets for the zoo are \$7 for adults and \$5 for children. When Justin went to the zoo with friends, the total cost of tickets for Justin and his friends was \$44. How many tickets did Justin and his friends purchase?

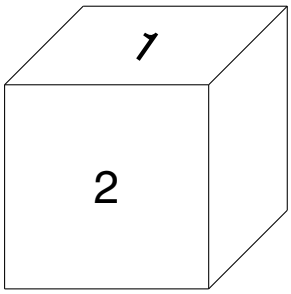
3. In the addition problem below, each of the digits 1, 3, 4, 5, 6, 7, and 8 replaces exactly one letter, and the resulting problem is correct. What is the product of the digits used to replace *E*, *F*, and *G*?

$$\begin{array}{rccc} & & A & B \\ + & & C & D \\ \hline & E & F & G \end{array}$$

4. Sandra looks at her digital watch and the time is 1 : 23 : 45, or 23 minutes and 45 seconds after 1 o'clock. What is the sum of the values for the hour, minute, and seconds on Sandra's watch 12345 seconds later?

5. Three years ago, the sixteen houses on Pi Street had an average age of twelve years old. That year, a number of new houses were built on Pi Street. No other homes have been built on Pi Street since then. Today, the average age of the houses on Pi Street is eleven years old. How many new houses were built on Pi Street three years ago?

6. As shown below, the top side of a cube is labeled with the number 1 and the front side of the same cube is labeled with the number 2. The remaining four sides are labeled with the numbers 3 through 6 so that consecutive numbers are always on sides that are next to one another. In how many different ways can the remaining four sides be numbered?



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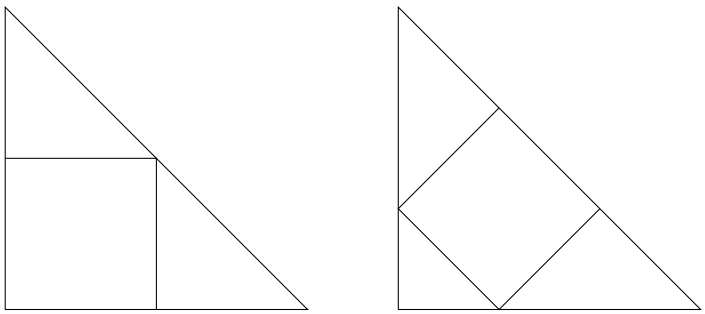
7. Tommy and Timmy are standing together on a circular running track. One lap around the track is 400 meters. Timmy and Tommy start running at the same time in opposite directions around the track. Timmy runs at a speed of 4.5 meters per second, and Tommy runs at a speed of 3.5 meters per second. When Timmy and Tommy meet for the fifth time, they stop running, and then together walk the shortest possible distance on the track back to where they started running. How far, in meters, did Tommy and Timmy walk together?

8. How many non-congruent obtuse triangles have whole number side lengths and a longest side of length 10?

9. Shown below are two congruent isosceles right triangles. A square is inscribed in each triangle. What is the ratio of the area of the smaller square to the area of the larger square? Express your answer as a common fraction.



10. The famous archeologist Dr. Jones has uncovered a small collection of ancient coins. Five of the coins are identical gold coins, two of the coins are identical silver coins, and one coin is bronze. On each of the coins, exactly one side has an engraving of the pharaoh. How many different ways can Dr. Jones stack the collection of coins into a single stack so that no two engravings of the pharaoh touch?

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Answers 12129

Sprint Round

1. D
2. A
3. A
4. B
5. A
6. B
7. A
8. D
9. D
10. D
11. E
12. A
13. B
14. D
15. D
16. A
17. D
18. A
19. D
20. A
21. C
22. C
23. E
24. C
25. E
26. A
27. B
28. C
29. D
30. B

Target Round

1. ($)486
2. 56
3. 5
4. 60
5. 6
6. 286
7. 27
8. 361

Team Round

1. 17
2. 8
3. 12
4. 83
5. 8
6. 10
7. 75
8. 12
9. $\frac{8}{9}$
10. 1512

Number Sense

1. 230
2. 2
3. 1302
4. 171
5. 169
6. 369
7. 156
8. 1
9. 28
10. $[3721, 4111]$
11. 946
12. 297
13. 196
14. 1
15. 3960
16. 1260
17. 496
18. 819
19. 1170
20. $[121676, 134482]$
21. $\frac{3}{11}$
22. 2675
23. 141
24. 15
25. 47
26. $\frac{5}{14}$
27. 441
28. 5.5
29. $\frac{3}{4}$
30. $[12791, 14137]$
31. 8019
32. 34
33. 57
34. 1221
35. 5609
36. 21
37. 825
38. 24025
39. 47
40. $[94008, 103902]$
41. 8
42. 2464
43. $25\frac{36}{37}$
44. 5
45. 2125
46. 420
47. 31
48. 39491
49. 125000
50. $[1455, 1608]$
51. 152
52. 100001
53. 9
54. 12978
55. 12
56. 8125
57. 30
58. 7488
59. 46
60. $[7328565, 8099991]$
61. $\frac{11}{16}$
62. 11988
63. 2021
64. 9261
65. 150
66. $43\frac{4}{9}$
67. 2021
68. 9310
69. $\frac{3}{22}$
70. $[3340, 3691]$
71. 15
72. 1060899
73. 2112
74. 855625
75. 36
76. 24
77. 63
78. 120900
79. 127
80. $[876, 967]$

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Solutions 12129

Sprint Round Solutions

1. The value of 99×123 is $123 \times 100 - 123$, or $12300 - 123 = 12177$. The sum of the digits is $1 + 2 + 1 + 7 + 7 = \boxed{18}$.

2. The number of points Andrew earned for correct answers was $16 \times 4 = 64$, and the number of points Andrew lost for incorrect answers was $3 \times 1 = 3$, so Andrew's score was $64 - 3 = \boxed{61}$.

3. If a number is even, then it is a multiple of 2. Any number that is a multiple of both 2 and 3 is a multiple of 6. Since $6 \times 8 = 48$, the number of even numbers less than 50 that are also multiples of 3 is $\boxed{8}$.

4. A hexagon has 6 sides, an octagon has 8 sides, a decagon has 10 sides, and a dodecagon has 12 sides. The total number of sides is $6 + 8 + 10 + 12 = \boxed{36}$.

5. Since Alan cannot be first, the possible orders are Beth, Candace, Alan, or Beth, Alan, Candace, or Candace, Alan, Beth, or Candace, Beth, Alan, for a total number of different orders of $\boxed{4}$.

6. The number 510 is 17×30. Additionally, $17 \times 4 = 68$. Therefore the number must be $510 + 68 = 578$, which has a tens digit of $\boxed{7}$.

7. Victoria has $16 + 3 = 19$ quarters, which have a value of $19 \times \$0.25 = \$4.75$. She also has $11 - 5 = 6$ dimes, which have a value of $6 \times \$0.10 = \$0.60$. Altogether, the total value of Victoria's quarters and dimes is $\$4.75 + \$0.60 = \boxed{\$5.35}$.

8. A distance of 4 miles at a speed of 12 miles per hour will take $\frac{4}{12} = \frac{1}{3}$ of an hour, and in minutes this is $60 \times \frac{1}{3} = \boxed{20}$.

9. The greatest common divisor of 60 and 90 is 30. Any number that divides both 60 and 90 is also a divisor of 30. The divisors of 30 are $1, 2, 3, 5, 6, 10, 15, 30$, for a total of $\boxed{8}$.

10. The area of one of the four rectangles is $300 \div 4 = 75$. Additionally, the length of the rectangle is 3 times the width. Therefore the length of the rectangle must be 15 and the height must be 5. The perimeter of the entire figure is $3 \times 15 + 5 \times 5 = \boxed{70}$.

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11. There are 365 days in the year 2021, and 7 days in one week, so the year 2021 has 52 weeks and 1 day. Therefore, December 31st is a Friday, as are December 24th, December 17th, December 10th, and December 3rd. If December 3rd is a Friday, then December 1st is a $\boxed{\text{Wednesday}}$.

12. Because there are 60 more fifth graders than fourth graders, and the ratio of fourth graders to fifth graders is $5:7$, the total number of fourth graders is $\frac{60}{7-5} \times 5 = 150$, and the total number of fifth graders is $\frac{60}{7-5} \times 7 = 210$. Because the ratio of third graders to fourth graders is $4:3$, the total number of third graders is $\frac{4}{3} \times 150 = 200$. The total number of third, fourth, and fifth graders is $150+210+200 = \boxed{560}$.

13. Figure *III* has area $7 \cdot 7 = 49$. The height of figure *II* is $\frac{30}{2} - 9 = 6$, so the area of figure *II* is $9 \times 6 = 54$. By the Pythagorean Theorem, the other leg of the right triangle in figure *I* is $\sqrt{15^2 - 12^2} = 9$, so the area of figure *I* is $\frac{1}{2} \cdot 12 \cdot 9 = 54$. Since both *I* and *B* have area *I*, the figure with the greatest area is $\boxed{I\&II}$.

14. If 75% are green, then 25%, or $0.25 \times 120 = 30$ are red. Since only green ornaments are removed, the number of red ornaments in the box does not change. For 60% of the ornaments to be green, 40% must be green, and 30 is 40% of $30 \div 0.40 = 75$. For the total number of ornaments remaining the box to be 75, the number of green ornaments removed must be $120 - 75 = \boxed{45}$.

15. There are two possibilities. For the first possibility, the number is between 17 and 23. Since $23-17 = 6$, and $6 \cdot \frac{2}{2+1} = 4$, the number is $17 + 4 = 21$. For the second possibility, the 23 is the halfway between the number and 17. Since $23 - 17 = 6$, the number is $23 + 6 = 29$. The sum of 21 and 29 is $\boxed{50}$.

16. There are 3 options for the one main course. Since there are 4 vegetable options and students must select 2, there are $\binom{4}{2} = 6$ options for the two different vegetables. There are 2 options for dessert. Altogether, the number of possible different lunches is $3 \cdot 6 \cdot 2 = \boxed{36}$.

17. There are three cases to consider. In the first case, the fifth number is less than 12, so the median is 12. Since the median and mean are equal, the sum of the five numbers is then $5 \cdot 12 = 60$. The sum of the four known numbers is $28+20+4+12 = 64$. This would require the fifth number to be negative, which is an impossibility. In the second case, the fifth number is greater than 20, so the median is 20. The sum of the five numbers is then $5 \times 20 = 100$, and the fifth number is $100 - 64 = 36$. In the third case, the fifth number is between 12 and 18. Let the fifth number be x. Then $64 + x = 5x$, and x is 16. The sum of the two possible values is $36 + 16 = \boxed{52}$.

18. The length of the path, in inches, is $8.25 \cdot 12 = 99$ inches. Using $\frac{22}{7}$ as an approximation of π, the circumference of the circle is $\frac{22}{7} \cdot 2 \cdot 3.5 \approx 22$ inches. Therefore, the number of revolutions the wheel makes is approximately $\frac{99}{22} = \boxed{4.5}$.

19. The least positive whole number requiring four digits in base 4 is 1000_4, which is $4^3 = 64$. The greatest positive whole number requiring four digits in base 4 is 3333_4, which is $4^4 - 1 = 255$. Similarly, the least positive whole number requiring three digits in base 6 is 100_6, which is $6^2 = 36$. The greatest positive whole number requiring three digits in base 6 is 555_6, which is $6^3 - 1 = 215$. The count of numbers where these two portions of the number line overlap is $215 - 64 + 1 = \boxed{152}$.

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20. The expression $xy + x + y + 1$ can be rewritten as $(x+1)(y+1)$, and the expression $xy - x - y + 1$ can be rewritten as $(x-1)(y-1)$. Therefore, $38\heartsuit 28 = (38-1)(28-1)$, which is 999, and $22\heartsuit 20 = (22-1)(20-1)$, which is 399. Finally, $999\heartsuit 399 = (999+1)(399+1)$, which is $\boxed{400000}$.

21. Let the three-digit number be ABC, where A, B, and C are distinct digits. Then the other possible numbers using the same three-digits are ACB, BAC, BCA, CAB, and CBA. When these six numbers are summed, the value is $200A + 200B + 200C + 20A + 20B + 20C + A + B + C$, or $222 \cdot (A + B + C)$. Therefore, the sum must be a multiple of 222, and the only answer choice that is a multiple of 222 is $222 \cdot 14 = \boxed{3108}$.

22. One trillion is 10^{12}. The number 8 is 2^3, and $2^{10} = 1024$, or a bit more than 10^3, so $8^{\frac{10}{3}}$ is a bit more than 10^3. Therefore, $\left(8^{\frac{10}{3}}\right)^4 = 8^{\frac{40}{3}}$ is a bit greater than 10^{12}. The smallest whole number that is a greater than $\frac{40}{3}$ is 14 so the answer is $\boxed{8^{14}}$.

23. If all 6 vertices are used, there is 1 non-congruent polygon. If all 5 vertices are used, there is also 1 non-congruent polygon. If 4 vertices are used, the 2 unused vertices can be adjacent to one another, have one vertex between them, or two vertices between them, for a total of 3 polygons. If 3 vertices are used, the three unused vertices can be consecutive, for 1 possible polygon, partitioned into two consecutive unused vertices and one unused vertex, for another 1 possible polygon, or partitioned so that no two unused vertices are consecutive, for another 1 possible polygon. Altogether, the total number of possible noncongruent convex polygons is $1 + 1 + 3 + 1 + 1 + 1 = \boxed{8}$.

24. Each team plays each of the other 5 teams twice, so that each team plays $5 \times 2 = 10$ games. For each of the 6 teams to play 10 games, there must be a total of $\frac{1}{2} \cdot 10 \cdot 6 = 30$ games, since each game always involves two teams. In each of the 30 games, exactly 2 total points are always awarded. Therefore, the total number of points in the league standings must equal $30 \cdot 2 = 60$. The five known point totals sum to $13 + 12 + 9 + 7 + 2 = 43$, so the the number of points earned by the first place team was $60 - 43 = \boxed{17}$.

25. Because the two triangles with areas of 12 and 16 share a common altitude to the lower right vertex of the quadrilateral, their bases, must be in the ratio of $\frac{12}{16} = \frac{3}{4}$. The triangle of unknown area and the triangle of area 28 share an a common altitude to the upper left vertex of the quadrilateral. The bases of these two triangles are the same bases as the other two triangles, so the areas must have a ratio of $\frac{3}{4}$, and the area of the unkown triangle is $28 \cdot \frac{3}{4} = 21$. Therefore the area of the quadrilateral is $21 + 28 + 16 + 12 = \boxed{77}$.

26. The prime factorization of 18 is $3^2 \cdot 2^1$, so the prime factorization of 18^5 is $3^{10} \cdot 2^5$. For the divisors of this number, each power of 3 from 0 through 10 is paired with each power of 2 from 0 through 5. Thus, for divisors which have a power of 2 equal to 2^0, there are $0 + 1 + 2 + \ldots + 10 = 55$ total powers of 3 in their product. Similarly for each other possible power of 2. The highest power of 3 that evenly divides the product is $3^{55 \cdot 6} = \boxed{3^{330}}$.

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27. For Nala's bag to contain three red marbles and three blue marbles, she must draw the exact same set of colors from Michelle's bag as Nala draws from her bag. For Michelle's selection, there are $\binom{6}{2} = 15$ possible outcomes, and $\binom{3}{2} = 3$ of these have two blue selected. If Michelle selects two blue, then her bag will contain 5 blue marbles and 3 red marbles, and of the $\binom{8}{2} = 28$ selections for Nala, $\binom{5}{2} = 10$ are selections of two blue marbles. Thus, the probability that Michelle draws two blue marbles from Nala and Nala draws two blue marbles from Michelle is $\frac{3}{15} \cdot \frac{10}{28} = \frac{1}{14}$. By symmetry, the probability that both draws are two red marbles is also $\frac{1}{14}$. For Michelle's selection there are $15 - 3 - 3 = 9$ draws where one marble of each color is selected. If this occurs, then Michelle's bag will contain 4 blue marbles and 4 red marbles, and $\binom{4}{1} \cdot \binom{4}{1} = 16$ of the 28 selections will be one marble of each color. Thus, the probability that Michelle draws one red and one blue marble from Nala and Nala draws one red and one blue marble from Michelle is $\frac{9}{15} \cdot \frac{16}{28} = \frac{12}{35}$. Altogether, the probability that Nala's bag contains three red marbles and three blue marbles is $\frac{1}{14} + \frac{1}{14} + \frac{12}{35} = \boxed{\frac{17}{35}}$.

28. The product can be considered in three portions. The first portion of the product are the units digits ending in 1, 3, 4, 6, 7, 8, and 9. There are 10 sets of these units digits, and the product of each set has a units digit of 8. The units digits of powers of 8 cycle in a pattern of $8, 4, 2, 6$, so the units digit of 8^{10} is 4. The second portion of the product are the multiples of 10. The product of 20 and 50 is 1000, and may be discarded. The units digit of the remaining product is identical to the previous units digit calculation, and is also 8. Finally, there is the portion of the product that is $12 \cdot 15 \cdot 22 \cdot 25 \cdot 32 \cdot 35 \cdots 92 \cdot 95$. This is equivalent to $6 \cdot 11 \cdot 16 \cdot 21 \cdots 41 \cdot 46 \cdot 30 \cdot 50 \cdot 70 \cdots \cdot 190$. The numbers with the units digits of 1 may be discarded. Group the remaining terms as $(16 \cdot 50 \cdot 150) \cdot (6 \cdot 26 \cdot 36 \cdot 46) \cdot (30 \cdot 70 \cdot 90 \cdot 110 \cdot 130 \cdot 170 \cdot 190)$. This has a first non-zero units digit equal to the units digit of $(4 \cdot 3) \cdot (6) \cdot (9 \cdot 9)$, which is 2. Therefore the first non-zero digit of 99! is the units digit of $4 \cdot 8 \cdot 2$, which is $\boxed{4}$.

29. Summing the five equations yields $a(x+y+z)+b(x+y+z)+c(x+y+z)+2(a+b+c) = 2022+(x+y+z)$. This can be written as $(a+b+c)(x+y+z)+2(a+b+c) = 2022+(x+y+z)$. Since $a+b+c = 2021$, $2021(x+y+z) + 2 \cdot 2021 = 2022 + (x+y+z)$, or $2020(x+y+z) = -2020$, so the value of $x+y+z$ is $\boxed{-1}$.

30. The ratio of segment AB to segment CB is $\frac{12}{16} = \frac{3}{4}$, and the ratio of segment XB to segment AB is $\frac{9}{16}$, which is also $\frac{3}{4}$. Therefore, triangle ABC is similar to triangle XBA. Let the length of AX be k. Then the length of segment AC is $\frac{4}{3}k$, and $k + \frac{4}{3}k + 7 = 49$, so $k = 18$ and the length of AC is $\frac{4}{3} \cdot 18 = 24$. The side lengths of triangle ABC are 12, 16, and 24, so the semiperimeter of ABC is $\frac{12+16+24}{2} = 26$, and by Heron's Formula, the area of ABC is $\sqrt{(26)(26-24)(26-12)(26-16)}$. The square of the area is $26 \cdot 2 \cdot 14 \cdot 10 = \boxed{7280}$.

Target Round Solutions

1. Jay works 2 hours each Tuesday, 2 hours each Thursday, and 5 hours each Saturday, for a total of 9 hours per week. In 6 weeks, Jay works a total of $6 \times 9 = 54$ hours. At \$9 per hour, the total amount Jay earns is $\$9 \times 54 = \boxed{\$486}$.

2. The vertical side on the right of the figure and the horizontal side on the bottom of the figure have total length $8 + 15 = 23$. The four vertical sides on the left of the figure must have a total length of 8. The horizontal sides on the top of the figure must also have a total length of 15, but the indentation of length 5 adds a length of $5 + 5 = 10$. The total perimeter of the figure is $23 + 8 + 15 + 10 = \boxed{56}$.

3. Since all even numbers are divisible by 2, only odd numbers need to be considered. Additionally, since $\sqrt{120} < 11$, and all factor pairs of a number must have one factor less than or equal to the square root of the number, the odd numbers between 100 and 120 only need to be checked for divisibility by the 3, 5, and 7. The odd multiples of 7 are 105 and 119. The odd multiples of 5 are 105 and 115, and the odd multiples of 3 are 105, 111, and 117. After discarding these numbers, the numbers 101, 103, 107, 109, and 113 remain, and all must be prime. Therefore, the number of prime numbers greater than 100 and less than 120 is $\boxed{5}$.

4. Let J be the number of jars. Then the number of beads is both $5J+5$ and $6 \cdot (J-1)$. Setting these two equal, $5J+5 = 6 \cdot (J-1)$, or $5J+5 = 6J-6$, so J is 11. Then the number of beads is $5 \cdot 11 + 5 = \boxed{60}$.

5. The four-digit number $ABCD$ is divisible by 11 if and only if $(B+D)-(A+C)$ is a multiple of 11. Because $9 + 8 + 6 + 0 = 23$, one of $B + D$ or $A + C$ must be 6, and the other must be 17. If the hundreds and units digits sum to 17, the possible numbers are 6908 and 6809. If the hundreds and units digits sum to 6, the possible numbers are 9086, 8096, 9680, and 8690. Altogether, the total number of possible four-digit numbers is $\boxed{6}$.

6. Consider a Sticks-and-Stones approach. Suppose there are 3 black pebbles and 12 white pebbles. Arrange 3 of the black pebbles and 2 of the white pebbles in an alternating fashion, like $BWBWB$. This creates four possibilities for the remaining 10 white pebbles. Some number could be placed before the first black pebble, some number between the first and second black pebble, some number between the second and third black pebble, and some number after the third black pebble. Once this is done, a possible arrangement might be $WWWWBWWWBWWBWWW$. The positions of the black pebbles correspond to numbers. In the possible arrangement, the numbers would be 5, 9, and 12. The number possible placements of the 10 extra white pebbles into the 10 possible locations is the number of possible selections of three numbers that are not consecutive. By Sticks-and-Stones, this is $\binom{10+4-1}{4-1} = \binom{13}{3}$, which is $\boxed{286}$.

7. Because $2 \times 3 \times 4 \times 5$ is greater than 72, Kevin's number, which is the least, must be 1. Therefore the product of Lara's, Maria's, and Nicole's numbers must be 72. If Lara's number is 2, then the product of Maria's and Nicole's must be 36. The possibilities are 3 and 12, and 4 and 9. If Lara's number is 3, then the product of Maria's and Nicole's must be 24. The only possibility is 4 and 6. Lara's number cannot be 4 since $4 \times 5 \times 6$ is greater than 72. The sum of all possible values of Nicole's number is $12 + 9 + 6 = \boxed{27}$.

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8. Let the vertical distance from the top edge of the square to the point be x, and the vertical distance from the bottom edge of the square to the point be y. Additionally, let the horizontal distance from the left edge of the square to the point be a and the horizontal distance from the right edge of the square to the point be b. Finally, let the fourth distance from the point to the vertex be K. Then by the Pythagorean Theorem, $x^2 + b^2 = 8^2$, $x^2 + a^2 = 12^2$, $y^2 + a^2 = 21^2$, and $y^2 + b^2 = K^2$. Summing the first and third equations yields $x^2 + y^2 + a^2 + b^2 = 8^2 + 21^2$, and summing the second and fourth equations yields $x^2 + y^2 + a^2 + b^2 = 12^2 + K^2$. Therefore $8^2 + 21^2 = 12^2 + k^2$, and K^2 is $8^2 + 21^2 - 12^2 = \boxed{361}$.

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Team Round Solutions

1. Because $4 \times 25 = 100$, there are 25 multiples of 4 less than 101. Any number that is a multiple of 4 and 6 is also a multiple of 12 and must be subtracted from this total. Since $12 \times 8 = 96$, there are 8 multiples of 4 and 6 less than 101, and $25 - 8 = \boxed{17}$.

2. If 1 adult ticket was purchased, then the total cost of the childrens tickets would be $\$44 - \$7 = \$37$, which is impossible, since childrens tickets are \$5 each. But if two adult tickets were purchased, then the total cost of the childrens tickets is $\$44 - \$14 = \$30$, and the number of childrens tickets is $\$30 \div \$5 = 6$. There is no other possible multiple of 7 that could be subtracted from 44 to leave a positive multiple of 5. Therefore the total number of tickets purchased was $2 + 6 = \boxed{8}$.

3. Any two two-digit numbers can only sum to a three-digit number with a hundreds digit of 1, so E must be 1. Additionally, since the digit 1 is already used and the digits 0 and 2 may not be used, the sum must be greater than 130. So no tens digits of one of the two-digit numbers can be a 3, nor can the tens digits together be 4 and 5, 4 and 7, or 5 and 6. The tens digit of the sum cannot be a 7 or 8, as no two two-digit numbers are great enough. Additionally, because the ones digit of the sum cannot be a 1, 9, 2, or 0, the ones digits of the two-digit number cannot together be 4 and 7, 3 and 8, 5 and 6, 6 and 4, 7 and 3, 4 and 5, or 3 and 6. This only leaves a limtied number of possibilities, several of which are valid. For example, $56 + 78 = 134$, or $87 + 56 = 143$. In all cases, the product of the digits of the sum is $1 \times 3 \times 4 = \boxed{12}$.

4. Since there are 60 minutes in one hour and 60 seconds in one minute, there are $60 \times 60 = 3600$ seconds in one hour, so three hours would be $3 \times 3600 = 10800$ seconds, leaving $12345 - 10800 = 1545$ seconds. The number of seconds in 25 minutes is $60 \times 25 = 1500$. Thus, the time 12345 seconds later is 3 hours, 25 minutes, and 45 seconds later. This is $1 + 3 = 4$ for the value of the hour. The value of the seconds is $45 + 45 - 60 = 30$, and, carrying this extra minute, the value of the minutes is $23 + 25 + 1 = 49$. Therefore the sum of the hour, minute, and second values is $4 + 30 + 49 = \boxed{83}$.

5. The average age of the 16 homes on Pi Street today is $12 + 3 = 15$, so the sum of the ages of these homes is $16 \cdot 15 = 240$. Let H be the number of new homes built three years ago. Then these homes add a total of $3H$ years to the total ages of all of the homes, and the total number of homes is $16 + H$. Therefore, $\frac{240+3H}{16+H} = 11$, or $240 + 3H = 176 + 11H$. Solving for H, the number of new homes built was $\boxed{8}$.

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6. If the side labeled 3 is on either of the 2 sides that is next to both the 1 and the 2, then there are 2 choices for the side to label 4, and then, regardless of which side is labeled 4, 2 ways to label the sides with 5 and 6. This is a total of $2 \cdot 2 \cdot 2 = 8$ possibilities. If the side labeled 3 is opposite the side labeled 1, then there are 2 choices for the side to label 4, and then the sides labeled 5 and 6 must follow in only 1 way. This is a total of $2 \cdot 1 = 2$ possibilities. Altogether, the number of possibilities is $8 + 2 = \boxed{10}$.

7. Every time Tommy and Timmy meet on the track, they collectively run a distance of 400 meters at a combined speed of $3.5 + 4.5 = 8.0$ meters per second. Therefore Tommy and Timmy meet every $400 \div 8.0 = 50$ seconds. The fifth time they meet, Timmy has run a total of $5 \times 50 \times 4.5 = 1125$ meters. Since every 400 meters places Timmy back at the starting location, the total distance Tommy and Timmy walk together is $3 \times 400 - 1125 = \boxed{75}$ meters.

8. There are two constraints to consider. Firstly, by the Triangle Inequality, the sum of the lengths of the shortest two sides must be greater than 10. Secondly, the side of length 10 must be opposite the obtuse angle, so by the Pythagorean Theorem test for obtuse angles, the sum of the squares of the other two side lengths must be less than $10^2 = 100$. The shortest possible length for the shortest side is 2. If the shortest side has length 2, then the second shortest side can only have a length of 9. If the shortest side has length 3, then the second shortest side can have length 8 or 9. If the shortest side has length 4, then the second shortest side can have length 7, 8, or 9. If the shortest side has length 5, then the second shortest side can have length 6, 7, or 8. If the shortest side has length 6, then the second shortest side can have length 6, or 7. Finally, if the shortest side has length 7, then the second shortest side can only have a length of 7. Altogether, the number of possible non-congruent triangles is $1 + 2 + 3 + 3 + 2 + 1 = \boxed{12}$.

9. Let the side length of the square with sides parallel to the legs of the triangle be 12. Then the area of the square is 144, and the legs of the right triangle are $12 + 12 = 24$. Let the side length of the other square be x. Then $\frac{x}{\sqrt{2}} + x\sqrt{2} = 24$, so x is $8\sqrt{2}$. The area of the square is $\left(8\sqrt{2}\right)^2 = 128$, and the ratio of the area of the smaller square to the area of the larger square is $\frac{128}{144}$, or $\boxed{\frac{8}{9}}$.

10. There are a total of $5+2+1 = 8$ coins. Considering only the material of the coins, there are $\frac{8!}{5! \cdot 2! \cdot 1!} = 168$ ways to order the coins. Now consider the orientation of the coins. Let H represent a coin oriented in the stack with its engraving upward, and T represent a coin oriented in the stack with its engraving downward. The stack cannot contain the sequence HT, as that would result in two engravings touching. Thus, the stack can contain 9 coins with H orientation, or, from the bottom of the stack upward, some number of coins with T orientation followed by the remainder of the coins with H orientation. This results in $1 + 8 = 9$ possible orientations for the coins in the stack, and the total number of ways to stack the coins is $168 \cdot 9 = \boxed{1512}$.

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